Exp no: 1
17/7/2026

Aim:

To execute basic linux and windows commands

Linux command:

Pwd - Displays the current working directory

ls - List files and folders in a directory

cd - changes the current directory

cp - copies files or directories

mv - moves or renames files

rm - Delete files

cat - Display the contents of a file

chmod - changes files or directory permission

windows CMD command:

dir - List files & folder in current dir

cd - changes the current dir

copy - copies files

move - moves files or folder

del - Deletes one or more files

Type - Display the content of a text file

cls - clears the command promt screen

ren - Renames a files or folder

Result:

The basic linux and windows were executed successfully and their respective outputs were verified

Exp no: 2
7/07/26

Learning and assignment of IP address manually to computers

Aim:

To learn how manually assign an ip address, subset mask, default gateway and DNS server to a computer and verify the network configuration

Procedure:

1) Open control panel -> Network and sharing center
2) select change adapter settings
3) Right click the active network adapter and select properties
4) Open internal protocol version 4 (TCP/IPV4)
5) selec use the following IP address
6) Enter the followeng values
   => IP address : 192.168.1.10
   => Subset Mask : 255.255.255.0
   => Default Gateway : 192.168.1.1
   => Preferred DNS Server : 8.8.8.8
7) click OK and Save the setting
8) Verify the configuration using the IP config command

Output:

```
IPV4 Address . . . . . 192.168.1.10
subnet mask . . . . . 255.255.255.0
Default Gateway . . . . . 192.168.1.1
DNS server . . . . . 8.8.8.8
```

Result:

Thus, the IP address was successfully assigned manually and the system displayed

Exp no:3
23/7/26

Study of different Types of network cables and Crimping of cable with RJ45 connection

Aim:

To study different types of network cables and perform Crimping of a UTP cable using an RJ45 connection

Material Required:

* UTP (cat 5e / cat 6) cable
* RJ45 connection
* Crimping Tool
* wire strippes
* cable cutter
* LAN cable testers

Types of Network cables

i) UTP (unshielded Twisted Pair)

* Most commonly used for Ethernet network

ii) STP (Shilded Twisted Pair)

* Has shielding to reduce electromagnetic interference

iii) Coaxial cable:

* used for cable TV and older networking systems

iv) Fibre optic cable

* Uses light to transmit data at very high speeds over long distances

Crimping Procedures:

1) Cut the cable to the required length
2) Strip about 2-3 cm of the outer insulation
3) Untwist and arrange the wires according to the T568B standard:
   * White / orange
   * Orange
   * White / Green
   * Blue
   * White / Blue
   * Green
   * Brown
4) Trim the wires evenly
5) Insert the wires fully into the RJ45 connector
6) Test the cable using a LAN cable Tester

Result:

The network cable was successfully crimped on RJ45 connector and Proper connectivity.

Ex No: 4

Implement Packet sniffing using Raw Socket in Python

Aim:

To implement a packet sniffer using new socket in python that captures and analyzes network packets by extracting basic IP header info.

Requirements:

* Python 3.x
* Linux or windows
* Root / administration

A packet sniffer is a program that captures packet Travelling across a network interface.

* Gates raw socket
* Capturing incoming packets
* Extracts IP headers
* Extracts info such as source IP destination IP, protocol & packet length

Procedure:

* Open a terminal or command prompt with root / admin privilges
* Install python 3 if it is not already installed
* Create a python file named packet sniffer.
* Save file

Ex No:5
24/7/26

Develope customized ping command to test server connectivity

Aim:

To develope and execute customized ping commands in windows to test server connectivity

i) Why do we use ping command:

* Ping command is used to test network connectivity b/w two devices, verify server

→ Key uses for:

i) Verify internet & Network Reachability

ii) Measure Network Latency

iii) Detect Packet Loss

iv) Troubleshoot DNS issues

v) Test Network stability

ii) What is the purpose of ping command:

The command is to check if a specific device or web server is reachable across a network, and to measure how long it takes for data to travel there and back. The ultimate purpose of ping command boils down to Three fundamental things: reachability, latency and reliability

Primary purpose:

i) Diagnostic Testing
   → Local Network
   → Internet
      Eg: ping 8.8.8.8

ii) Measuring Network speed & Latency

iii) Detecting connecting Drops & unstability

iv) Isolating DNS problems

iii) Customized ping command to check connectivity

ping google.com

Description:

→ send ping request to the Google server
→ Checks whether the server is reachable
→ Display the response time and packetless

Output:

pinging googl.com [142.250.183.14] with 32 bytes

Reply from 142.250.183.14 bytes = 32 times = 48 ms TTL = 117

Reply from 142.250.183.14 bytes = 32 times = 20 ms TTL = 117

Reply from 142.250.183.14 bytes = 32 times = 19 ms TTL = 117

Reply from 142.250.183.14 bytes = 32 times = 18 ms TTL = 117

Q.
21

ping statistics for 142.250.183.14:

packets : sent = 4 , Received = 4 , Lost = 0

approximate round trip times :

Minimum = 18ms , Maximum = 20ms , Average =

Result :

The ping command was successfully studied and used to less network connectivity

Exp No:9

Implementation of subnetting in Cisco packet Transfer

Aim:

To implement classes IP is CISCO packet Tracer & verify command between different subnets

Procedure

1) Open cisco packed Tracer & create a new network topology
2) Add required devices such as routers, switches and Pcs to workspace
3) Connect devices using appropriate cable routers interfaces
4) use network 192.168.1.0/24 & divide it into smaller subnets using 9/127 subject task (255.255.224)
5) A/27. Subnet provides 30 usable host
6) 192.188.1.0/27
7) 192.168.2.0/27
8) 192.168.3.0/27
9) 192.168.4.0/27.
10) Configure routes - interface Through the cot Enter

Ex
24

Enable
Configure terminal
Enterface Fast Ethernet 0/0
IP add. 192.186.1.1.255.255.255.224
no shutdown
Exit
interface Fast Ethernet 0/1
IP address 192.168.2.1.255.255.255.224
no shutdown
Exit

Configure the switch port in ouers mode
Enable
Configure terminal
interface Fast Ethernet 0/1
Switch port mode ouers
Exit
Enterface fast Ethernet 0/2
Switch port mode ouers
Exit

1) Configure each Pc by selecting Pc → default → IP config. Enter appropriate IP address. Submit Task & default gateway

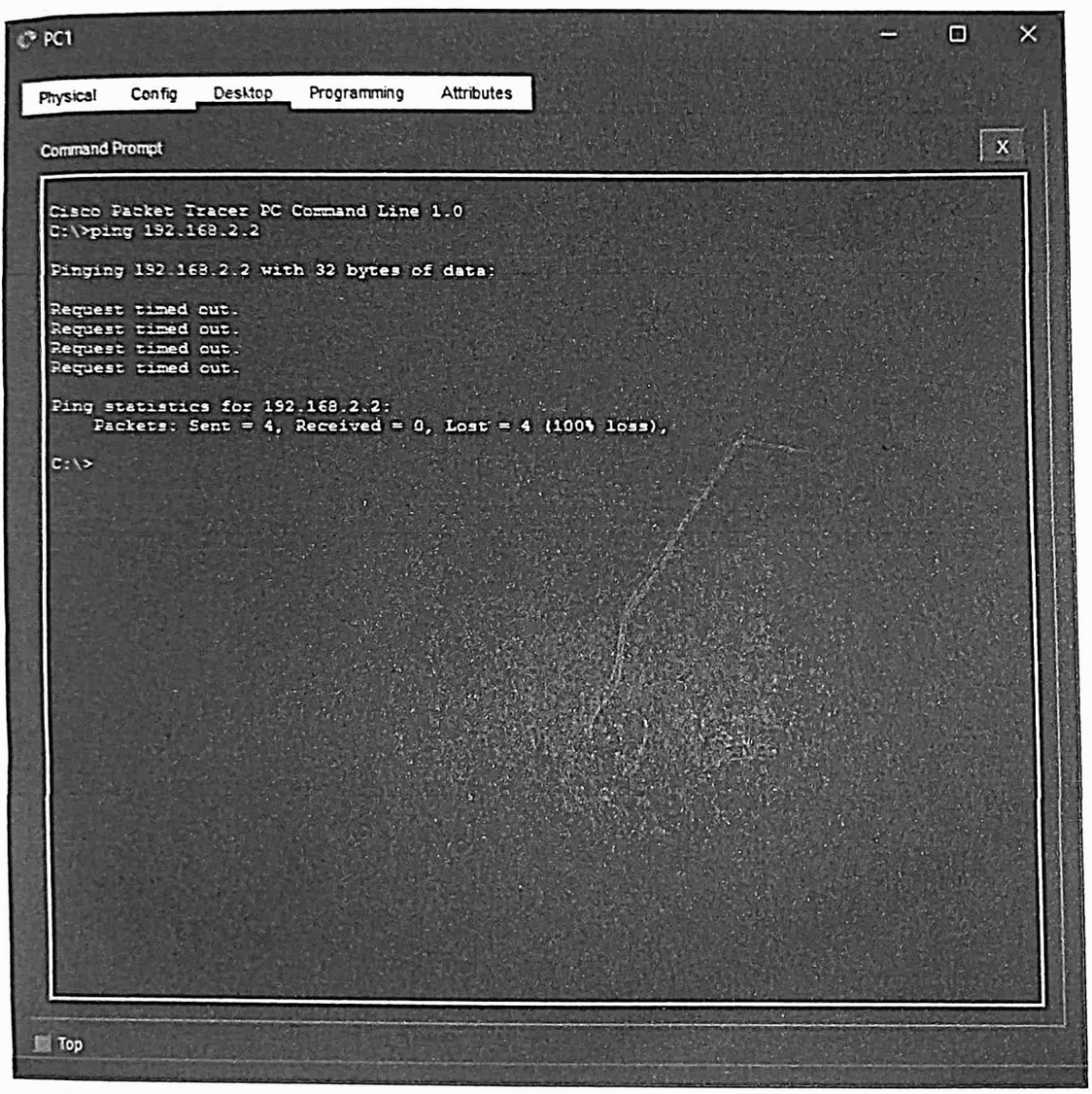
Observation:

a) understanding of subnetting

Subsetting is process of dividing a Large IP. network into smaller logical network

b) Advantage of Implementing subsetting

...tion of IP address



...bnetting

...ork to

Result:

Thus, the implementation of subnetting the Cisco packet Tracer executed successfully

Ex No:10

Customize switch with network module using cisco packet Tracer

Aim:

To Learn how to add a network module to a cisco switch. connect multiple PC's and a server, assign IP address and test connectivity in cisco.

Procedure:

1) open cisco packet Tracer and setup your workspace
2) Drag & drop a 2960-2477 switch to centre
3) Add four PC's & one Service - Pi from the End device pannel.
4) Power off switch
5) Install HWIC-HESw module on switch
6) Power on switch
7) Connect all devices using cooper straight through cable

Pc0 - switch Fast Ethernet 0/1

Pc1 - switch Fast Ethernet 0/2

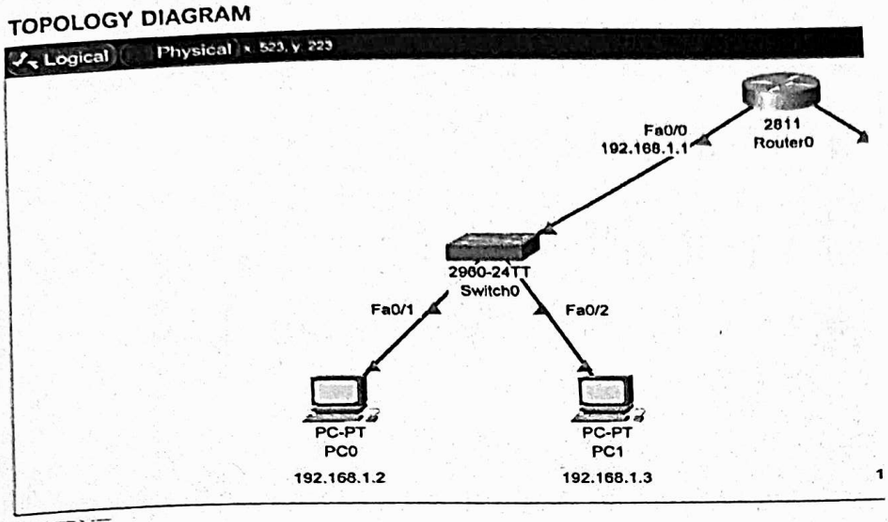
PC_2 - Switch module port 1

PC_3 - Switch module port 2

P_{24} - Switch module port 3 ... source

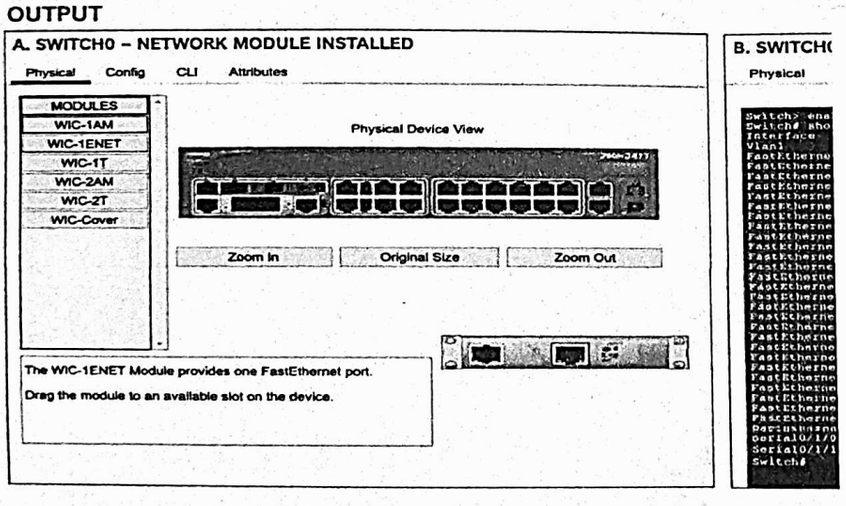
EXPERIMENT 10: CUSTOMIZE SWITCH WITH NETWORK ...

TOPOLOGY DIAGRAM



...2960-24TT Switch

OUTPUT



...which ...fully ...module ports

Result:

The switch was successfully customized with a network module, and all connected devices communicated successfully.

Ex No: 7

Develope A simple calculation using XML RPC

Aim:

To develope a simple calculator using XML-RPC protol in python that perform Addition operation like addition, subtraction and division through client server communication

Description:

XML-RPC is a protocol that is XML to encode request and response and HTTP as a Transport mechanism It enable communication between a client and a server where the client invokes method host on the server immediately

- Server to host arithmetic function
- client connect to server to perform arithmetic operation

Software requirement

- OS: window / Linux
- python version:
- Model required: XML RPC . server . XML . RPC . client
- Text editor or IDE : IDLE / VS code / Py charm

Procedure:

- Open python IDLE terminal
- create two seperate python server py and client py
- Run server . py file first to start XML RPC
- Observe the result and verify server output

client side code:

```
import xmlrpc.client
proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")
print("Add: 10+5", proxy.add(10,5))
print("Sub: 10-5", proxy.subtract(10,5))
print("mul: 10*5:" proxy.multiply(10,5))
print("Division: 10/5 =" Proxy.division(10/5))
print("Division by zero 10/0 =", proxy.division(10,0))
```

O/P:

Add : 10 + 5 = 15

Sub : 10 - 5 = 5

multiply : 10 x 5 = 50

Division : 10/5 = 2.

Division by zero : 10/0 = Error : Division by zero is not allowed.

Result:

The XML-RPC based calculator was successfully implemented the client communication with the server using XML-RPC, performed all arithmetic operation and displayed accurate result

Ex Exp No:12

Nmap To Discover Live Host Using ARP Scan and TCP/UDP ping Scan in TryHackMe platform

Aim:

To discover live hosts in a network using different Nmap host discovery Technique such as ARP Scan, ICMP ping, TCP ping and UDP ping on the TryHackMe platform.

Requirements:

- TryHackMe account
- Nmap
- Terminal
- Target IP address

Command:

1) Basic Host discovery

```
nmap -sn <target>
```

2) ARP Scan

```
Sudo nmap -sn -PR <target>
```

3) TCP SYN Ping

```
Sudo nmap -sn -PS <target>
```

4) TCP SYN ping on specific Ports

Sudo nmap -sn -PS22,80,443 <target>

5) TCP ACK ping

Sudo nmap -sn -PA <target>

6) TCP ACK ping on specific Ports

Sudo nmap -sn -PA80,443 <Target>

7) UDP ping

Sudo nmap -sn -PU <target>

8) UDP ping on specific ports

Sudo nmap -sn -PU 53,161 <target>

9) ICMP Echo ping

Sudo nmap -sn -PE <target>

10) ICMP Timestamp Ping

Sudo nmap -sn -PP <target>